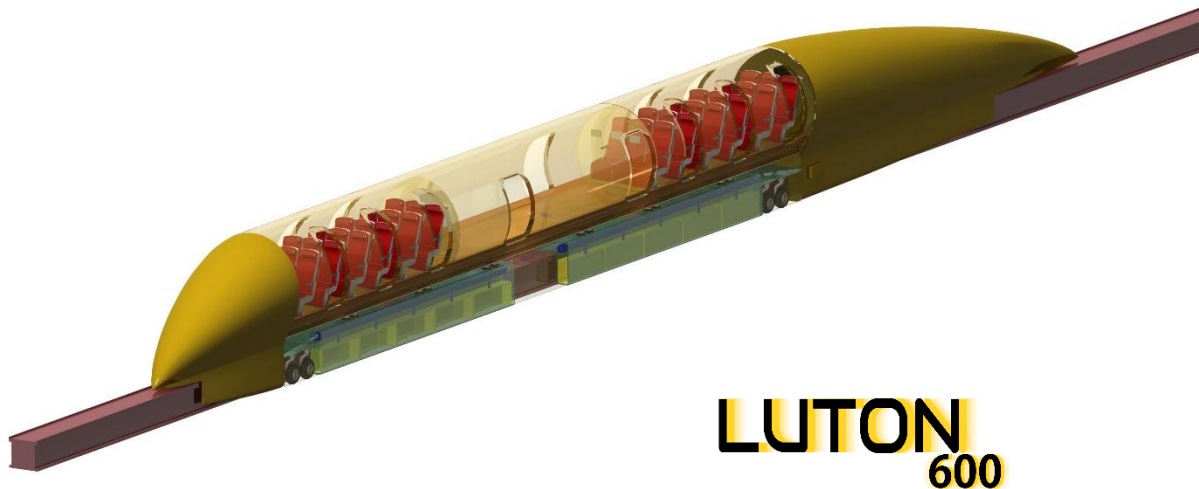


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Aerodynamic Design

Individual Report



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Abstract

This report presents the aerodynamic design of the Vacuum Tube Express, an evacuated tube, U.K. based, transportation system designed as a direct contender to regional jets. A review of current systems is presented and the superiority of the proposed concept is outlined. The aerodynamic design drivers are identified and their impact on drag is analysed using a 2D axisymmetric CFD study. Theoretical and numerical flow choking considerations are explored and the choice of 0.249 blockage ratio between capsule and tube is reasoned. The idea of using a compressor to push the blockage ratio up is briefly explored in the form of a cost breakdown, showing clearly the advantages of having one, however the associated design complexity prevents us from incorporating a compressor at this stage. The flow regime is explored using CFD and identified as transitional, with a LSB forming under the APG on the tail of the capsule. The aerodynamic shapes and fineness ratios for the nose and tail were optimised with regard to drag considerations. The 2D axisymmetric assumptions were tested when full 3D simulations of our system setup were run and found to not reflect the exact flow physics. Longitudinal vortices developed due to flow spillover from the magnets' air gap and their effect on the aerodynamic forces and moments, as well as transition and separation was investigated. Lastly, the final design is presented and areas for improvement and further investigation are singled out, while an objective discussion is carried out on whether streamlining the body is advantageous or not for such a system.

Acknowledgements

It is interesting to note how, perhaps, one of the largest deterrents to human knowledge is complacency, and how with every advance that we make, we build larger walls for ourselves to climb by resting on our laurels. A suitable metaphor portraying the human quest for knowledge would be that of a mountain climber, who as soon as he sees the summit, takes a ‘well deserved’ rest, thinking that his goal is within reach. Unbeknownst to him is the fact that the summit is a mirage, and that his path is never-ending. But all is not lost, since there are those who break down the walls of complacency, both for themselves and for others, and I was lucky to meet such a person in my supervisor, Dr. Paul Bruce, to whom I am grateful for his dedicated guidance and constructive criticism.

I am also grateful to my colleague in the Aerodynamics team: Marcin Rywik for his innovative ideas and contribution to a great working environment. My special thanks go to my other teammate and long-lasting friend Alexandru Dorcioman for his tireless dedication throughout this project, and his refusal to give up, even when no solutions were foreseeable. Needless to say, none of the work presented here would have been possible without the contribution of all members of the Aerodynamics team, and all members of this Group Design Project.

Many thanks are also due to Dr. Chris Cantwell, who granted us access to Imperial’s Aeronautics Computational Cluster (not to mention that without his teachings we would not have been able to find our way around a Linux Cluster).

Most of all, I would like to thank my parents for being there for me when I needed it the most, and helping me through my toughest times with their love, compassion and understanding.

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Nomenclature

$\dot{m}_{choked}$	Choked mass flow rate (kg/s)	M	Mach number
γ	Heat capacity ratio, for air 1.4	M_∞	Upstream Mach number
γ	Intermittency factor ($\gamma = 0$ -laminar $\gamma = 1$ -turbulent)	M_{ch}	Choking Mach number
i, j, k	Cartesian unit vectors	M_{crit}	Critical Mach number
ω	Specific turbulence dissipation rate 1/s	M_{DD}	Drag divergence Mach number
ρ	Density (kg/m ³)	P	Power (W)
τ	Blockage Ratio $\tau = \frac{A_{capsule}}{A_{tube}}$	p	Pressure (Pa)
θ	Momentum Thickness (m)	Re_{θ_t}	Transition Onset Reynolds number based on θ
ε	Turbulent dissipation (J/(kgs))	Re_H	Reynolds number based on hydraulic diameter
A^*	Throat area where $M = 1$ in a Laval nozzle	Re_L	Reynolds number based on length
$A_{capsule}$	Cross-sectional area of the capsule (m ²)	S	Reference area (m ²)
A_{tube}	Cross-sectional area of the tube (m ²)	T	Temperature (K)
C_D	Dimensionless drag coefficient	Tu	Turbulence intensity
C_f	Skin friction coefficient	U	Velocity (m/s)
C_L	Lift Coefficient	V	Volume (m ³)
C_M	Pitching Moment Coefficient	W	Work (J)
C_P	Pressure Coefficient	y^+	Dimensionless wall distance
D	Drag (N)	APG	Adverse pressure gradient
$D_{capsule}$	Capsule cross-sectional diameter (m)	BL	Boundary Layer
D_H	Hydraulic Diameter (m)	c.g.	Center of Gravity
D_{tube}	Tube cross-sectional diameter (m)	CPR	Compressor pressure ratio
k	Turbulent kinetic energy (J/kg)	DNS	Direct Numerical Simulation
L	Total train length (m)	LES	Large Eddy Simulation
l_m	Prandtl's mixing length (m)	LSB	Laminar Separation Bubble
L_N	Nose length (m)	MoC	Method of Characteristics
L_T	Tail length (m)	RANS	Reynolds Averaged Navier Stokes equations
		SPL	Sound Pressure Level (dB)
		SST	Shear Stress Transport
		URANS	Unsteady RANS

Introduction

At a time when air transportation reigns supreme over long distances, it is hard to imagine that the way forward could be a ground (or underground) transportation system. Nonetheless, there have been people in the past century who have identified the potential of an evacuated tube concept, starting with simple speculation in 1910, and continuing with the first concrete idea put forward by Salter ([1]) and the first proper feasibility study performed by Forgacs ([2]). The idea has recently been revived by Space X in the form of the “Hyperloop” ([3]) and gained considerable popularity, but many key technical aspects were left unexplored. This report aims to investigate a potential low-loss, evacuated tube, magnetic suspension, transportation system as a direct contender to jet aircraft. Such a system should, in theory, exhibit superiority over rival systems in air and sound pollution, conservation of energy resources, ticket costs, safety and reliability and ultimate speed without compressibility-induced effects. Air-transport, meanwhile, is expected to still be superior in its capability of serving low-traffic density areas and overseas routes and its flexibility in circumventing equipment failures ([2]).

Objectives

The main objectives of the aerodynamic design are: choking prevention and blockage ratio selection, selection of tube pressure level and pumping system design, assessment of system pressure sensitivity, capsule aerodynamic optimisation, prediction of flow transition, investigation of 3D flow effects, and estimation of noise, forces and moments acting on the capsule. This individual report will aim to: provide a review of current ground transportation systems and propose an alternative to regional jets in the form of evacuated tube transportation, outline the methods employed in the preliminary study and identify the design drivers, describe the flow choking mechanism and obtain the required blockage ratio, study the impact of a compressor on cost reduction, explore the possibility of maintaining laminar flow and its impact on drag, optimise the aerodynamic shape via an axisymmetric study, and finally investigate the 3D effects of the flow, their importance and present the final design choice. This report together with my colleagues’ contributions will address comprehensively

the aerodynamic objectives and assess the overall aerodynamic feasibility of such a concept.

Review of Current Systems

When it comes to high-speed ground transportation systems, the conventional choices are: high-speed rail or Maglev. Both are well-researched and proven to work, but the power required to overcome aerodynamic drag at high velocities becomes exorbitant ($P \propto U^3$), and the aerodynamic noise levels provide serious issues ($SPL \propto U^6 - U^8$). Moreover, the aerodynamic issues which need to be overcome do not stop here: in the open, moderate crosswinds can lead to increases of up to 60% in drag ([4]), while also affecting lift, side forces and impacting stability (the rolling moment being particularly sensitive to changes in yaw angle). In addition, tunnels arise as the solution to the desire of having straighter tracks for high-speed operation, while also being required due to environmental considerations in densely populated areas. Aerodynamically speaking, tunnels lead to the generation of pressure waves and an increase in drag: for long tunnels the increase in drag is dominant, while for short tunnels the most noteworthy effects (especially in sound levels) come from the pressure pulses generated at tunnel entry and exit ([5]). Moreover, the highly unsteady pressure history leads to transient loads and passenger discomfort and the streamwise pressure gradients affect separation and reattachment near the nose and the tail. Current Maglev designs capable of reaching up to 580km/h require careful streamlining to keep drag and noise levels within acceptable limit, while also minimising lift. Due to the aforementioned reasons, it is difficult for high-speed rails, or Maglev systems alone to provide a contender to regional jets in terms of journey time.

Proposal

A solution for higher velocities is identified in a partially evacuated tube, where the low drag compared to open atmosphere brings forth the promise of low-losses and the magnetically levitated propulsion system leads to low air pollution. The need to streamline the train’s body depends on the initial pressure, but should not have a capital effect. The optimum pressure to which the tubes should be evacuated is affected by pumping

costs as a function of pressure, propulsion costs as a function of drag and changes in the technical properties of air with pressure ([2]). Moreover, the sound attenuation of the low-pressure confined environment should permit the system to operate quietly for system users and neighboring non-users alike. Our proposal lies in a U.K. based high-speed ground system which will rival regional jets as far as velocity and travel times are concerned (unlike HS1 [6] and HS2 [7]). The system, named VTX (Vacuum Tube Express, [8]), will be launched in three phases: Phase 1 London-Luton, Phase 2 London-Manchester and Phase 3 London-Glasgow. This report, although presenting the general aspects of the system, will focus on the design of Phase 1. The proposed solution consists of two partially evacuated tubes in which capsules transporting about 30 people per capsule will travel. Each journey will take approximately 6 minutes with a cruise velocity of 600km/h. The capsules will be magnetically levitated using Halbach arrays, while superconducting magnets are used for lateral stability and linear synchronous motors (LSM) are employed for propulsion. Most of the track/tube will be overground and will follow the M1 motorway between London and Luton, while near the two cities, the track will be underground for environmental reasons.

Feasibility

In the conceptual stages of the design, a very important question to answer is whether the system is feasible or not. From an aerodynamic standpoint, this reduces to finding the power requirements of the pumping system, and consequently fixing the tube pressure that would provide an optimum between the pumping costs and costs associated with overcoming drag. In order to be able to do so conservatively, drag was estimated using a 1D Unsteady MoC analysis ([9]), treating each capsule as a piston, while the work required to evacuate the tube and maintain the vacuum (overcoming potential leakage, [10]) was estimated using the fact that: $W = \int p dV$. Assuming a daily functioning time of 20h and a mean of 28 capsules passing through each tube per hour, an optimum pressure of $p = 110\text{Pa}$ was found. This provides us with a starting point around which to design, but as our drag, pumping and leakage estimates become

more accurate this pressure is bound to change and should be updated in future iterations.

Preliminary Design

The aerodynamic design of such a concept (train in a partially evacuated tube) is still under current research ([11], [12], [13], [14], [15], [16], [17], [18], [19]) but still no overarching, universally-accepted conclusions have been drawn as to the actual advantages (if any) streamlining the body has, or what are the key aerodynamic aspects which should be investigated if such a project is to be developed. Consequently, the decision was taken that a study particularly tailored towards our application should be initiated within the aerodynamics team. The aims of such a study are: identifying the key design drivers, as well as the main factors which affect the flow inside a partially evacuated tube, developing an understanding of the flow regime, the 3D effects, and optimising our design accordingly.

Methodology

A dimensional analysis would indicate that the flow over a capsule inside a tube would be dictated by 3 dimensionless parameters: Re_L , Re_H , and M . Assuming a temperature inside the tube of $T = 298\text{K}$ (close to atmospheric conditions), and a cruise velocity of 166.67m/s we find that $M = 0.482$, meaning that compressibility effects cannot be neglected. Given that the density is so low ($\rho \propto p$ from the ideal gas law) it follows that the Reynolds numbers are low as well (compared to open atmosphere). This fact makes CFD a powerful tool applicable even in the incipient stages of such a design. The commercial software STAR-CCM+ ([20]) was employed in order to obtain solutions to the compressible Reynolds Averaged Navier Stokes (RANS) equations, initially employing a fully turbulent $k-\epsilon$ model (the inadequacy of this choice will be addressed in the Transition section). Our initial setup concerned an axisymmetric body placed in the middle of the tube, where, in order to capture the physics of a capsule travelling through initially stationary air, a change of frame of reference was employed and the tube walls were modelled as moving viscous walls. A depiction of a usual computational domain can be seen in Figure 13. For a more comprehensive discussion

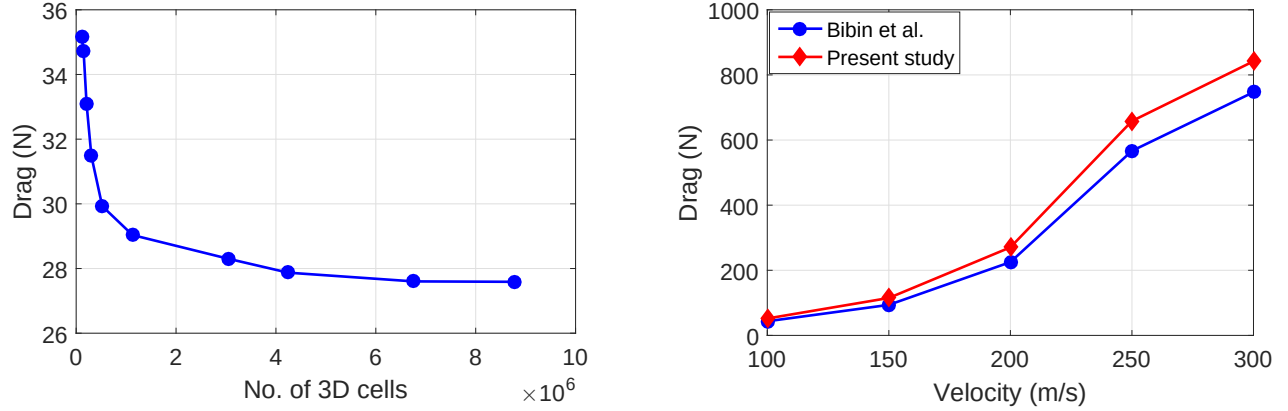


Figure 1: Validation of the numerical methods employed. On the left a mesh convergence study for a blockage ratio of 0.13 is highlighted, while on the right, drag values from the current study are compared against data obtained by Bibin et al. ([19]) for a blockage ratio of 0.21.

on the numerical setup (boundary conditions, mesh quality etc.) please see Appendix A. In order to validate to some extent our initial setup, the following steps were taken: check the dependence of values of interest on domain size (no significant changes were observed as long as the flow was not choked), do a mesh convergence study, check against previously published results. A mesh convergence study was performed for the baseline model presented in Figures 13 and 14 at a blockage ratio of 0.13 (not choked) and our cruise conditions. It is clear from Figure 1 that as the mesh was refined the total drag converged towards a value of 27.5N. Moreover, the baseline model presented in reference [19] was then used (spherical-cap front, flat back, $L = 40\text{m}$, $p = 1013.25\text{Pa}$), and as velocity was varied the drag obtained in the present study was compared in Figure 1 against the drag valued obtained by Bibin [19]. The trends are very similar but the values do not match exactly and this can be explained most likely by the fact that, in their paper, the exact inlet temperature is not provided, hence the exact M_{inlet} is not known. A comparison was drawn against the data presented in [19] since in that study a solution was sought for the compressible Navier-Stokes equations (in other references such as [11], [12], [14], incompressible flow is assumed for Mach numbers of up to 0.88, which in reference [19] was found to be inapplicable for velocities higher than 150m/s, due to the large difference in drag).

Design Drivers

The next step was to identify and assess the contribution of the major aerodynamic design drivers. One key parameter is undoubtedly velocity, or better put in this context, Mach number. The usual variation of drag is quadratic with velocity, up to the onset of compressibility effects. The flow over the body reaches sonic conditions at $M_\infty = M_{crit}$ and there is a rapid rise in drag at $M_\infty = M_{DD}$. However, in this particular context, as the capsule travels faster through the tube, a point will be reached when the flow is choked and there will be a high pressure build-up in front of the capsule, leading to a sharp rise in drag. This will be detailed in the Choking section. However, as long as the flow is not choked, the low drag values of a capsule in a partially evacuated tube do not pose restrictions on velocity. The other two parameters which play a key role in the design of such a concept were identified as being the pressure inside the tube and the blockage ratio. For the pressure study, the same baseline model was kept all throughout (cylinder with spherical caps, as shown in Figure 13) and the blockage ratio was kept fixed at 0.25, with $D_{capsule} = 2.4\text{m}$ and $D_{tube} = 4.8\text{m}$; Mach number was fixed at $M_\infty = 0.482$, and $T = 298\text{K}$ for inlet conditions. Evacuated tube pressures ranging from 100Pa to 100,000Pa were investigated and the results are shown in Figure 2. Note that as p increases, Re also increases, and in order to achieve convergence for the highest pressures, the mesh had to be refined progressively. Both the variation of the pressure drag and the skin friction

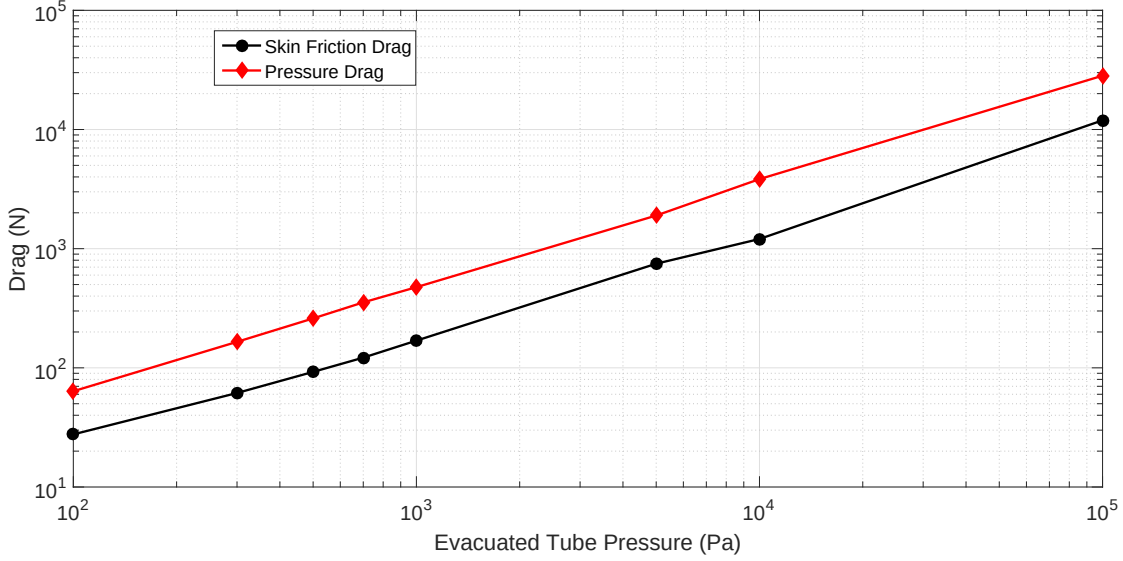


Figure 2: Variation of pressure drag and skin friction drag with pressure inside the tube presented in log-log axes for clarity.

drag with pressure inside the tube are very close to linear in log-log axes which suggests a power series dependence of the drag on tube pressure. The slopes of both lines are close to 0.9 which suggests a near-linear variation, while the ratio between pressure drag and skin friction drag stays around 2.5. The near linear variation can be explained by considering the fact that:

$$D = \frac{1}{2} \rho U^2 S C_D \quad (1)$$

and that, from the ideal gas law, $\rho \propto p$. The high pressure to skin friction drag ratio can be explained by the high curvature at the tail induced by the spherical cap design which encourages early separation and a considerable-in-size turbulent wake characterised by a large zone of recirculation, as seen in Figure 3. The blockage ratio study ([21])

was undertaken by keeping $p = 110\text{Pa}$ fixed, while varying τ between 0.1 and 0.8. The Mach number and the temperature were kept the same as before and the main conclusions can be summarised as follows: as τ increases the pressure difference between the front and back of the body increases, and hence the pressure drag increases, while as the blockage ratio increases, velocity over the contraction increases, causing an initial increase in skin friction drag, but as the flow chokes and the mass flow remains constant, velocity does not change much, and the decrease in C_f with Re leads to the skin friction drag plateauing. The consequence is that the ratio between pressure drag and skin friction drag increases with increasing blockage ratio. This analysis however is steady, and does not capture the full story of flow choking, which is a highly unsteady phenomenon ([22]).

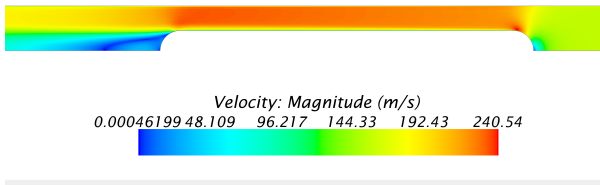


Figure 3: Velocity variation for the baseline model with a blockage ratio of 0.2, illustrating a typical turbulent wake with a large recirculation zone, which develops due to the spherical cap design at the tail which promotes early flow separation with no possibility of reattachment. Note that flow is from right to left.

Design Limitations: Choking

What actually dictates when the flow around a high-speed train in a partially evacuated tube chokes, and what are the consequences of that? Since the flow is subsonic (even in a high subsonic regime the same considerations apply), the comments in the Hyperloop concept ([3]) that flow choking is dictated by the Kantrowitz limit were found to not be applicable in the context of this study. The Kantrowitz limit ([23]) is related to

supersonic diffusers, and the deceleration of air from **supersonic** velocities in channels, where a normal shock in the diverging part of the diffuser is necessary for stable flow. Consequently, this dictates that the maximum contraction ratio is equal to the contraction ratio required to isentropically compress the flow to $M = 1$ (from initial supersonic conditions) multiplied by the total head ratio across the normal shock. Since the capsule is travelling at subsonic speeds, the analysis which would be more appropriate in this particular case is that which involves choking of subsonic channel flow ([22], [24]).

Theoretical Aspects

When flow approaches the body, it is initially accelerated on the body surface up to $M = 1$, and then, across the sonic line, the flow is further accelerated, leading to a supersonic flow pocket. At the end of this region a recompression brings the flow back to a subsonic state. As M_∞ is increased, the recompression shock becomes stronger and more extended towards the walls of the tube; if the shock reaches the tube's walls, the sonic line also extends all the way to the walls, and in this case the flow Mach number cannot be increased because the maximum mass flow is reached, and hence the flow is choked. To illustrate this graphically, consider Figure 4. The supercritical flow shown in Figure 4a retains the essential features of subsonic flow, an arbitrary perturbation influences the entire flowfield, although the mechanism of propagation is different in the infinite subsonic region and the finite supersonic region ([25]). As M_∞ increases, the supersonic pocket grows in size and it becomes infinite when $M_\infty = 1$ as seen in Figure 4b, where the sonic line extends to infinity (S), and so does the shock wave emanating at the trailing edge (T). For a supersonic stream ($M_\infty > 1$) the subsonic region becomes finite and is contained between the head shock wave and the sonic line, as seen in Figure 4c. In the case of a confined flow, however, a pattern very similar to that of sonic conditions, occurs well before M_∞ , at choking conditions, when the sonic line extends all the way to the wall, as shown in Figure 5. We can estimate inviscidly the choking Mach number for a given blockage ratio by starting from the Laval nozzle

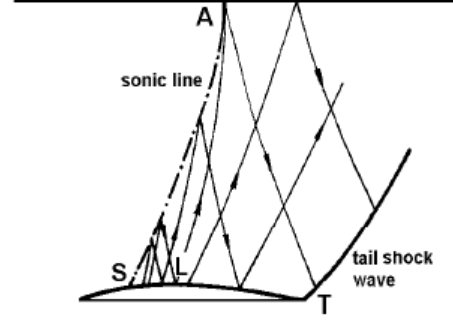


Figure 5: Subsonic channel flow choking, depiction from reference [24].

isentropic area ratio:

$$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{(\gamma+1)/(\gamma-1)} \quad (2)$$

and the fact that our blockage ratio would simply be:

$$\tau = 1 - \frac{A}{A^*} \quad (3)$$

because A^* would be the annular area between the tube and the capsule. We can rearrange to arrive at ([24]):

$$\tau = 1 - M_{ch} \left[1 + \frac{1}{6} (M_{ch}^2 - 1) \right]^{-3}. \quad (4)$$

The dependence of τ on M_{ch} given by equation (4) is known as the Allen-Vincenti parabola ([24]), and admits two solutions: a subsonic M_{ch1} in which we are interested, and a supersonic M_{ch2} which physically requires the maximum Mach number in the subsonic region of Figure 4c to be below M_{ch1} . So if we set $M_{ch1} = 0.482$ (our M_∞) we find a corresponding $\tau = 0.273$. Now the question arises, how do we correct this result in order to account for viscous effects, such as BL growth?

Numerical Aspects

As far as choking in such situations is concerned, a steady CFD solver behaves in a very similar way to an actual wind tunnel ([24]): if $M_\infty < M_{ch1}$ then the true Mach number is kept as the inlet Mach number, but if $M_{ch1} \leq M_\infty \leq M_{ch2}$ then the code overrides the specified inlet Mach number (to the closest choked Mach number) by invoking a compression of the upstream layers of the mesh. This is

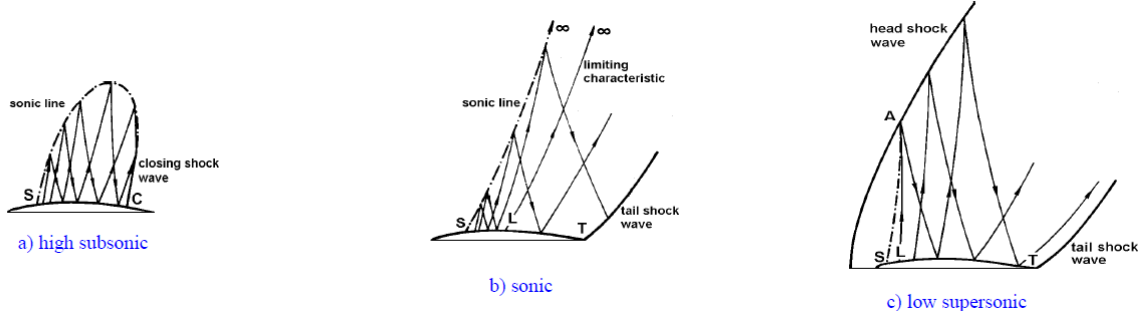


Figure 4: Comparison between expected compressible flow regimes of an airfoil in free-stream. Image found in reference [24].

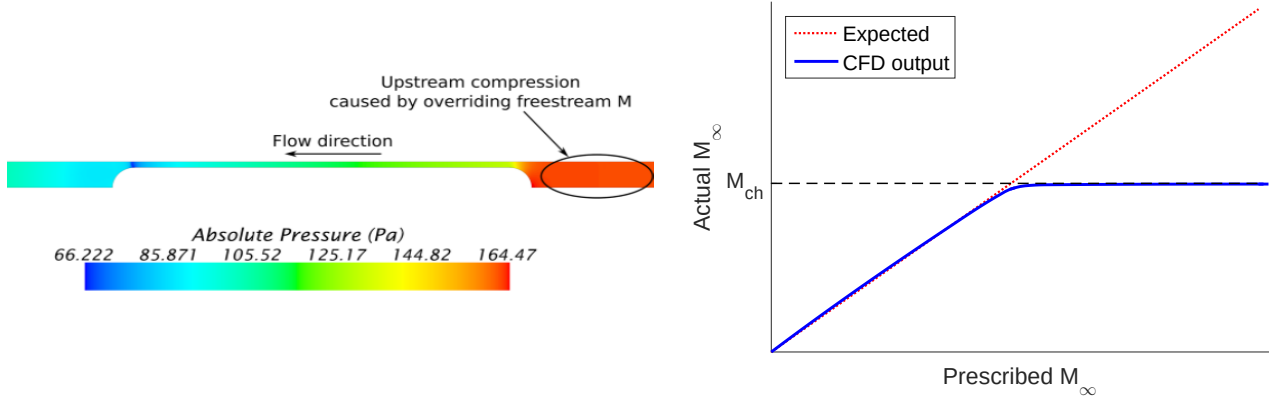


Figure 6: Choked subsonic channel flow and the numerical behaviour of the CFD solver employed. The left side image illustrates static pressure for $\tau = 0.6$, while the right side image illustrates qualitatively the numerical choking described.

exemplified in Figure 6, where an artificial pressure build-up caused by overriding M_∞ is visible on the left, while on the right, a qualitative sketch of the solver's behaviour is shown. The reason behind this is that subsonic channel flow choking is an unsteady phenomenon. This was illustrated in reference [22] where pressure measurements on a choked wind tunnel model were found to be highly transient. This is also exemplified in Table 1, where a comparison between a steady (RANS) and an unsteady (URANS, not capable of capturing fully an unsteady flow, but capable of illustrating fluctuations in the mean flow quantities) choked flow simulation is drawn.

Table 1: Steady vs Unsteady results for $\tau = 0.6$, baseline model.

	Steady	Unsteady
Pressure Drag (N)	312.27	371.34
Skin Friction Drag (N)	43.23	43.98
Maximum M	1.15	1.3

It can be noticed that skin friction drag is not really affected by the change, but the maximum Mach number rises in the unsteady simulation, while there is also an over 20% increase in pressure drag. Throughout the blockage ratio study, when we compared the CFD overridden M_∞ to the inviscidly predicted M_{ch1} we noticed that the former was lower in value, and we identified the cause for that to lie in the extent of the viscous effects: BL growth. As the boundary layer grows, the effective area that the flow sees becomes smaller, hence ($\tau_{viscous} > \tau_{inviscid}$) and the flow is choked at a smaller M_∞ than predicted. Based on the values observed for the choked cases in the blockage ratio study, an average correction to $\tau_{choking}$ of 3% was applied. Hence, for $M_\infty = 0.482$ the desired blockage ratio is corrected from 0.273 to 0.265. To account for lower temperatures during winter, hence higher Mach numbers corresponding to speeds close to 600km/h the decision was taken to set $\tau = 0.249$, for robustness. These values are

clearly quite restrictive, and in order to attempt to push the allowable blockage ratio up, the idea of a compressor at the front, to unchoke the flow ([3]) was investigated.

Compressor Considerations

The possibility of incorporating a compressor into our design was analysed as follows: isentropic relations were used to determine $\dot{m}_{choked}$, and based on a given blockage ratio, the mass flow rate required to unchoke the flow was found. This analysis was inviscid, and in order to account for BL growth a compressor and an exhaust nozzle were modelled simplistically in the CFD solver as a pressure outlet, where pressure and temperature were prescribed in accordance to the desired mass flow rate, and a velocity inlet, respectively, where the velocity and temperature of the jet were specified based on compressor performance. This analysis allowed us to obtain reliable mass flow rates and compressor inlet areas as a function of blockage ratio, and it allowed the propulsion team to find the compressor pressure ratio (CPR) required, again, as a function of τ . For a detailed account of the calculations see [21]. In order to determine whether a compressor would be a viable solution or not, a cost estimation was initially performed in conjunction with the propulsion and guideway and station design teams, where the major cost drivers which varied with blockage ratio were considered: compressor costs, tube and tunneling costs, and pumping costs. The compressor costs were split into: capital costs, maintenance costs, and power costs, and were considered for a total of 16 capsules. The capsule diameter was fixed at 2.4m which meant that the tube diameter varied with τ which incurred serious material and tunneling costs, as well as pumping costs required to maintain the vacuum. The results of the cost analysis for a span of 20 years are presented in Figure 7. The operational costs for the compressor on the right of Figure 7 are very similar irrespective of τ and this is explained by the fact that the CPR does not vary much with τ (between 2 and 3). The cost conclusions presented in Figure 7 are quite convincing, and the need for a compressor would be even more evident in Phases 2 and 3 where the Mach numbers will be higher. However the decision was taken to **not** incorporate a compressor in the

current design due to the following reasons: the compressor inlet was designed such as to be close to choking, hence the piping area requirements (we required the flow through the pipes not to be choked) to connect the compressor to the exhaust nozzle were too large. In order to decrease the area required for the pipes, the possibility of increasing CPR was considered. If an axial compressor would be used this would result in too many stages and too much complexity (and a considerable decrease in cost savings), and if a centrifugal compressor would be used this would result in an impeller area larger than the cross-sectional area of the capsule, hence without further R&D the idea of having a compressor was discarded. For a more in-detail discussion of the decision-making involved, please see [9].

Aerodynamic Optimisation

Since no compressor was incorporated in the design, the blockage ratio was kept at the value previously mentioned of $\tau = 0.249$, and the idea of streamlining the body was pursued in order to determine the extent of the drag reduction possible and its impact.

Flow Regime

Even though a fully turbulent $k-\varepsilon$ model was used so far in the analysis, we ask ourselves whether we would expect the flow to be fully turbulent, and the answer is no. If we assume a capsule length $L = 30\text{m}$ it follows that $Re_L = 3.46 \times 10^5$ which is below the usually quoted value for transition onset of a flat plate (5×10^5). We also expect the fact that the confinement accelerates the flow to provide a favourable pressure gradient ($\frac{dp}{dx} < 0$) and help delay transition ([26]). However, there are many more factors at play as far as transition to turbulence is concerned ([26]), such as: free-stream turbulence, $\frac{dp}{dx}$, Re , M , acoustic radiation, surface roughness, surface temperature, surface curvature and the history of these parameters. All these factors make determining the onset and end-of-transition points a very difficult task. Mathematical stability theory (the Orr-Sommerfeld equation and envelope e^N method, [27]) is applicable to good extent but it still requires numerical solutions and it is known to not give accurate predictions

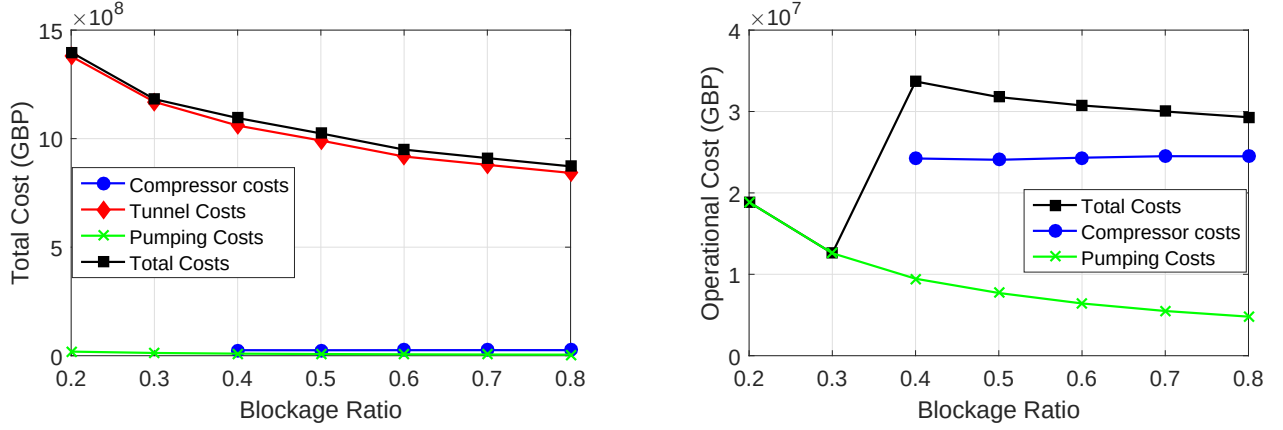


Figure 7: Total and operational costs over 20 years indicating clearly that cost-wise a compressor is desired to push the blockage ratio up and reduce total material costs for the tube and tunneling costs.

for complex geometries, where three dimensional effects are present, hence the most accurate tools to predict transition to date remain experiment ([26]), LES ([27]) and DNS ([28]).

CFD for Transition

Naturally, none of the options mentioned above are available to us, hence in order to obtain at least qualitative results, STAR-CCM's $k - \omega \gamma - Re_{\theta_t}$ model was employed. This model, initially proposed by Menter ([29]) is a low- Re model (based on SST $k - \omega$), correlations based, where two extra transport equations are used: for γ and for Re_{θ_t} . Two vital correlations in Menter's model are proprietary, hence the model implemented in STAR-CCM+ and used throughout this study is the one presented in [30]. In order to ensure better results, the mesh was refined as suggested in [31] and the free-stream edge function required to distinguish between free-stream and BL was also implemented as advised in [31]. In order to validate our understanding of the model, as well as the effect of confinement and free-stream Tu , a flat plate study was undertaken and is presented in [9]. The effect of confinement, was as expected, providing a favourable pressure gradient and delaying transition, while the effect of Tu was as described in [26] with variations of C_f ranging from fully laminar (Blasius) to transitional and fully turbulent. In the particular case of the geometries we are interested in, with our usual Re and M the physical mechanisms of transition were investigated and are presented in [21]. The main

findings can be summarised as follows: for the total lengths under consideration (between $L = 26.4\text{m}$ and $L = 48\text{m}$) the flow was observed to remain laminar all the way up to the tail of the capsule, where, under an adverse pressure gradient ($\frac{dp}{dx} > 0$), it separated and transitioned through a laminar separation bubble (LSB), as also observed for this range of Re in [27]. Another common finding with reference [27] was the fact that, as the Re increased (by increasing tube pressure), the LSB was reduced in size, in our case this decrease in size was found to be near-exponential. It was also found that there was a near-exponential trend in the rate at which the end-of-transition point moved closer towards the front of the body as Re was increased. These findings stand to show us how the transition point and mechanisms would be affected by a higher/lower tube pressure or cruise velocity for the capsule. The effect of varying M was however not investigated.

Shape Optimisation

Using the transitional model we went on to optimise the aerodynamic shape in the same 2D axisymmetric context. Please note that for this study it was assumed that $D_{capsule} = 2.4\text{m}$. Several shapes were initially tried with the $k - \epsilon$ model ([9]) and three of them were chosen for the transitional study: LD Haack, power series ($x^{1/2}$), and modified superellipse. The LD Haack series is designed to minimise drag for given length and diameter (this is indeed our case since diameter will be fixed by the pressure vessel and we set the length by

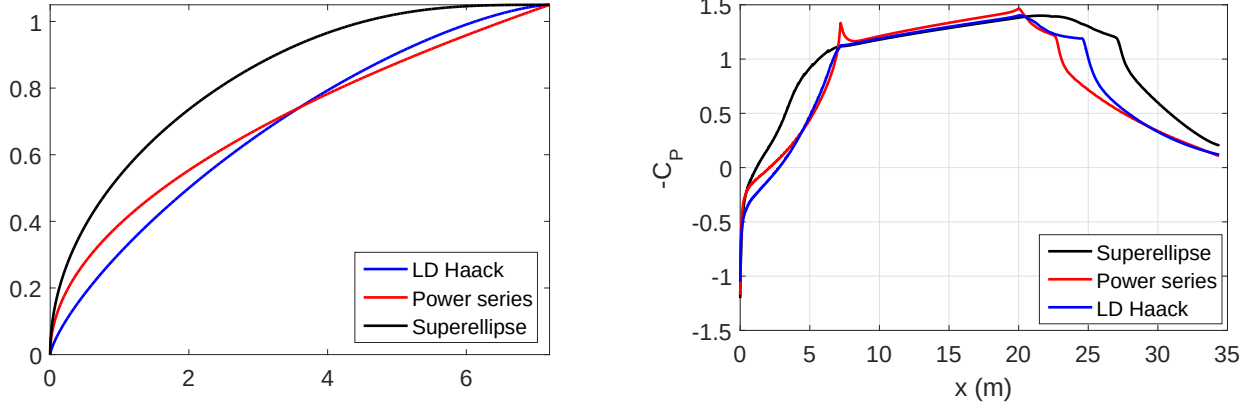


Figure 8: Shape study using transitional CFD model. The total length was fixed and the nose and tail fineness ratios were fixed at 3 and 6, respectively. The shapes mentioned in the legend were used for both nose and tail.

varying the fineness ratio) and it is also known to be superior in transonic flow (good for Phases 2 and 3) with a continuous gradient at the junction with the body. The power ($x^{1/2}$) series is also known to be superior in transonic flows and have good pressure recovery performance, however it has a discontinuous gradient at the joint between the nose and body, which affects BL receptivity and encourages transition to turbulence. The superellipse ([32]), on the other hand, eliminates the leading-edge-flat-plate juncture discontinuity, eliminates curvature discontinuity, but moves the pressure recovery region closer to the leading edge, which means a poorer pressure recovery performance. The three shapes are plotted next to each other in Figure 8, along with the C_P distribution on the capsule surface for each design. The results of the shape study were tabulated and are presented in Table 2, for a fixed total length of 34m with a nose fineness ratio of 3 and a tail fineness ratio of 6. It can be noticed that the transitional model provided significantly lower C_D values due to the laminar flow (it is also interesting to note how the percentage of total drag due to skin friction changes from turbulent to laminar). Between the three shapes discussed, we notice that the power series has the highest percentage of skin friction, followed closely by the LD Haack, and then the superellipse with the lowest. This follows what we see on the right of Figure 8, where the transition point (corresponding roughly to the middle of the LSB) is marked by a discontinuity in $\frac{dC_P}{dx}$ on the tail: power ≈ 23 m, haack ≈ 24 m, superellipse ≈ 27 m. We can notice in Table 2 that the design with

power series both at nose and tail is slightly better than the LD Haack, however the decision was taken to have a LD Haack shape, both at nose and tail, because it was very close to the best option, and we believed that the RANS model employed would not accurately capture BL receptivity due to the discontinuous gradient at the junction of the power series. Hence we judged that the power series design could transition considerably earlier, hence the safer option (Haack) was chosen.

Fineness Ratio Optimisation

Once the length of the pressure vessel was fixed by the structural team (19.2m), and the shape was chosen, a study was undertaken to fix the length of the nose and the tail, and hence their fineness ratios. This study was undertaken solely using the transitional model, and unfortunately, due to time restrictions, not as many configurations as initially planned were studied. However an optimum was still found, since increasing the fineness ratio, especially at the tail, reduces the chances of separation, and hence reduces pressure drag. However, increasing length, increases wetted area, hence skin friction drag. The optimum configuration was found to be for a nose fineness ratio of 1.5 ($L_N = 3.6$ m), and a tail fineness ratio of 4.5 ($L_T = 10.8$ m), and this is illustrated in Table 3. The C_f and C_P distributions for the final design choice can be seen in Figure 9, where the main characteristics of transition to turbulence are also highlighted. Once this choices have been made, the decision was taken to move to a three dimensional

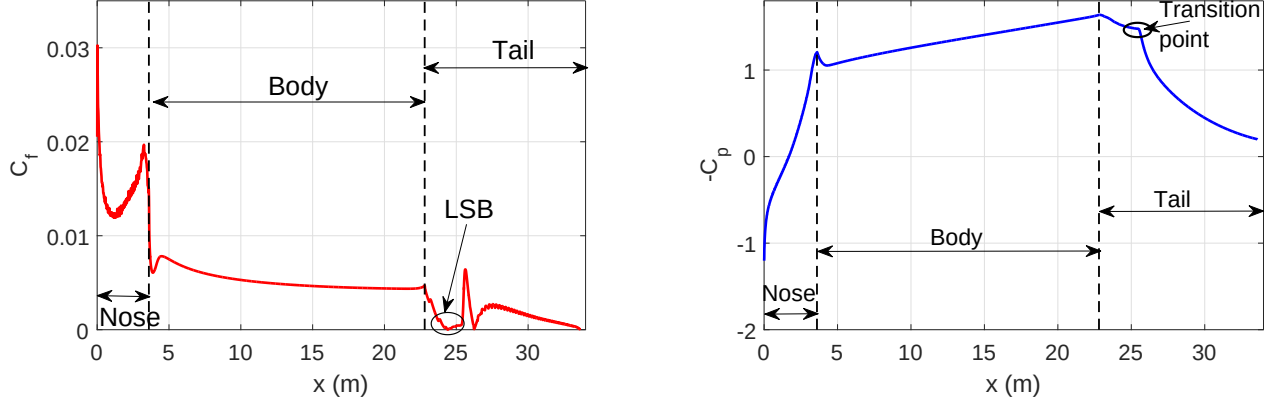


Figure 9: C_f and C_p distributions for the final design choice using LD Haack for both nose (fineness ratio 1.5) and tail (fineness ratio 4.5) are shown. The main transition characteristics are also shown in the form of the laminar separation bubble and the discontinuity in $\frac{dC_p}{dx}$ over the tail.

analysis and study the impact of: ground effect, gap between magnetic track and capsule, vertical position of nose tip and tail tip etc. The decision was taken to fix the nose downwards, such as to direct as much flow as possible over the capsule, and not in the small gap between the magnetic track and the underside of the capsule.

Three Dimensional Effects

A 3D CFD study was undertaken using the same transition model, solving the three dimensional, steady, compressible RANS over half the body/domain in order to exploit the geometrical symmetry and reduce computational costs. According to the relative frame of reference used throughout, the whole tube was modelled as a moving viscous wall, and now also the column-shaped magnetic track (see Figure 15 or 16) was modelled as a moving viscous wall. The length of the computational domain upstream and downstream of the capsule was kept the same as in Figure 13 with the same inlet and outlet conditions. The study was focused around 2 different geometrical shapes: tail down (Figure 17) and an axisymmetric tail (Figure 18) and 3 gap (between magnetic track and capsule) distances: no gap, 3.5cm (set by Propulsion) and 10cm. Due to the extensive time demands of the simulations (each simulation took an average of 20h on the Aeronautics Cluster to converge to desirable accuracy), only 4 configurations were studied: no gap-tail down, 10cm gap-tail down, 3.5cm gap-tail down/up. For a discussion on the meshing and its considerable challenges for this setup, please see

Appendix D. The effect of varying the air gap will be discussed in detail in reference [21] while this report will focus on the optimisation of the tail position for the 3.5cm gap and comparing the 3D results against the 2D axisymmetric predictions. Irrespective of the tail position, the physics observed in the 3D simulations were very different to what the 2D axisymmetric simulations predicted. Similar behaviour as observed in reference [33] (which was later confirmed experimentally, [34]), was noticed in the present study. Firstly, the flow passing through the air gap, due to the small area and high pressure build-up, is pushed out and spills over, leading to the formation of longitudinal vortices along the side of the capsule. This is illustrated on the left of Figure 10, while the fact that the vortices move away from each other in the wake of the body due to ground effect is shown on the right. The formation of these vortices is illustrated by looking at crossflow velocity vectors at various streamwise positions in Appendix B. It was noticed that the larger the gap, the stronger and larger the side vortices were. Hence for the 10cm gap-tail down the flow did not separate over the back of the body due to the two side vortices, while for the 3.5cm it still separated, however the location of separation was pushed back as compared to the no gap simulation. The axisymmetric tail design displayed no separation over the back (as predicted in the 2D study), but its major shortcoming was in the large area of recirculation on the underside of the tail. This came as a consequence to the low mass flow rate

Table 2: The main results of the shape optimisation study, showing the marked difference between fully turbulent and transitional model, hence highlighting the need to consider a transitional flow regime. The C_D values given are for total drag with S being taken as the cross-sectional area of the capsule. For the drag values, the percentage of pressure drag and skin friction drag is also shown.

Nose	Tail	$k - \varepsilon$			$k - \omega \gamma - Re_\theta$		
		C_D	Pressure (%)	Skin Friction (%)	C_D	Pressure (%)	Skin Friction (%)
Haack	Haack	0.846	53.95	46.05	0.599	61.95	38.05
Super	Super	0.966	54.43	45.57	0.674	65.63	34.37
Power	Power	0.826	53.15	46.85	0.597	61.33	38.67
Super	Haack	0.885	54.31	45.69	0.666	68.62	31.38
Haack	Super	0.931	54.34	45.66	0.705	70.68	29.32

Table 3: The main results of the fineness ratio optimisation study, highlighting a drag optimum for a nose fineness of 1.5 and tail 4.5.

Fineness Ratio		C_D	Pressure (%)	Skin Friction (%)
Nose	Tail			
1.5	1.5	0.941	81.65	18.35
1.5	3	0.745	74.21	25.79
1.5	4.5	0.724	70.89	29.11
3	3	0.785	74.25	25.75
3	6	0.736	68.32	31.68
4.5	4.5	0.768	70.34	29.66

from under the capsule compared to the high mass flow rate over the tail, leading to flow spilling over at the tip of the tail. The differences in velocity fields for the two configurations are illustrated in Figure 11, while the pressure distributions are compared in Figure 12. The performance of the two configurations is also tabulated and presented in Table 4. We can judge by the considerable increase in C_D that there is a need to consider three dimensional effects in the design, which is also unavoidable if one desires relatively accurate values for C_L and C_M . It is also clear from Table 4 and Figure 12 that the tail down design was better in terms of minimising both drag and lift, and hence, it was chosen as this iteration’s final design. There is also a need to consider 3D optimisation in the regards that the 2D optimum shape might not be the 3D optimum, as was found in [33]. It is interesting to note that the proportion which skin friction drag takes out of the total drag also increases, and this is due to the higher C_f values on the underside of the body than on the top. What is also interesting is the fact that for the tail down simulation (3.5cm gap), the temperature on the

lower side of the capsule reached values up to 660K, but we have no way of knowing if this accurately reflects the actual maximum temperatures, since the BL, and hence the thermal boundary layer was not well captured on the underside (see Appendix D). For the 10cm gap the maximum temperature did not exceed 326K, however as the gap size is decreased we expect the temperature to increase, but to what extent, should be the subject of further investigation.

Discussion

It has often been said, that for an evacuated tube system, there is no need to actually streamline the body ([1], [2], [11], [12]). Looking in hindsight on our design, however, it is hard to take one stand or the other. On the one hand, the 2D study showed that streamlining the body reduced drag to half its original value, but such a saving might easily not apply to the 3D case due to the large difference in flow physics. Preliminary calculations done in conjunction with the Business team and the Propulsion team showed that

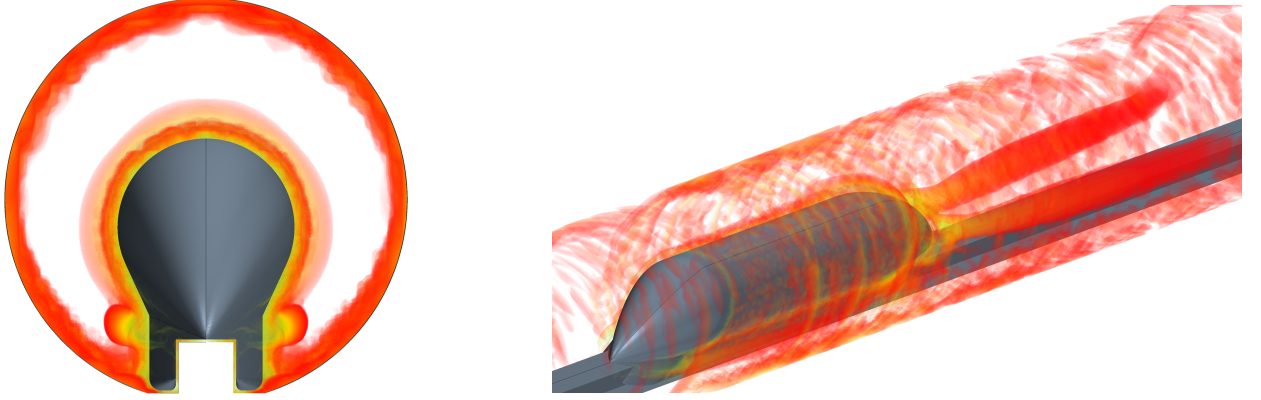


Figure 10: Volume rendering of the λ_2 Criterion showing the longitudinal vortices forming due to air-gap spill-over and moving away from each other due to ground effect.

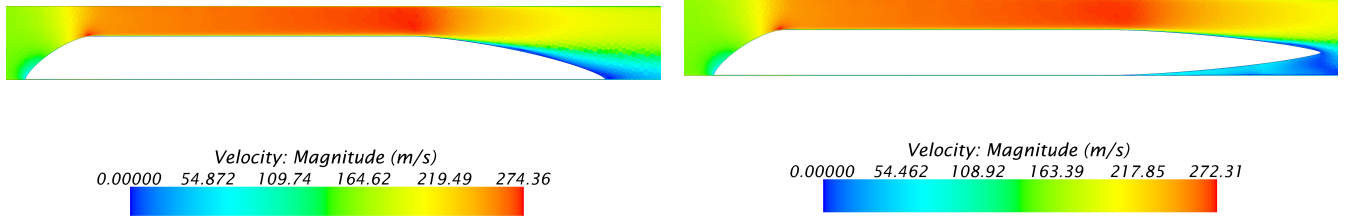


Figure 11: Velocity fields for the tail down and axisymmetric tail configurations. Slice taken $z = -0.01$ (spanwise location). Flow is from left to right.

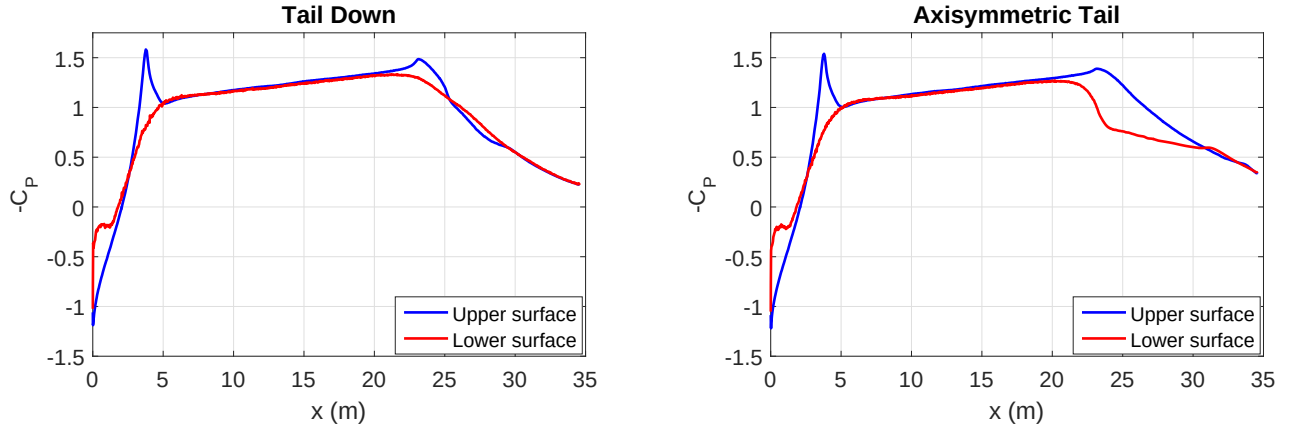


Figure 12: C_P distributions for the tail down and axisymmetric tail configurations. They both exhibit similar trends, however the tail down design has better pressure recovery at the back, and the lower surface trend is different between the two at the tail, due to the recirculation zone of the axisymmetric tail design.

Table 4: The main results of the 3D optimisation study are compared against the 2D predictions, showing the need to consider three dimensional effects. Note that length chosen in non-dimensionalising the pitching moment was the capsule length, $L = 33.6\text{m}$ and the point about which the pitching moment was taken is the c.g. ($x = 13.35\text{m}$ from nose tip streamwise, $y = 1.31\text{m}$ from lower side of the wheels).

	C_D	Pressure (%)	Skin Friction (%)	C_L	C_M
Tail Down	0.964	62.49	37.51	0.158	0.008
Axisymmetric Tail	1.05	66.92	33.08	0.523	0.171
2D prediction	0.724	70.89	29.11	-	-

the energy required to overcome 1N of drag per journey is 0.0167kWh, while the energy required per journey to accelerate/decelerate etc. 1kg of mass is 0.0067kWh. Hence consulting with the Structures team and considering the extra 158.68kg added for the nose and tail, we reach the conclusion that this would be worth it if we could save over 63.66N of drag. Considering our final drag value of 77.9N ($C_D = 0.964$), and assuming that the same savings from 2D apply, streamlining the body would be justified, not to mention the added aesthetic appeal which would help marketing the product. Also streamlining the body will have more of an effect in case of unwanted leakage and increase in tube pressure. However, this analysis does not include manufacturing costs, or maintenance costs, so overall, it is hard to say.

Final Design and Remarks

The main aerodynamic design parameters and features are summarised in Table 5..

Table 5: Summary of the main features of the aerodynamic design of the capsule.

Total Capsule length	33.6m
Nose length and fineness ratio	3.6m 1.5
Tail length and fineness ratio	10.8m 4.5
Blockage Ratio	0.249
Cruise Velocity	600km/h
Design Mach number	0.482
Cross-sectional area	5.05m ²
C_D	0.964
C_L	0.158
C_M	0.008
Tube pressure	110Pa
Aerodynamic Noise	62dB
Nose shape	LD Haack
Tail shape	LD Haack
Air Gap	3.5cm
Proportion of laminar flow on upper and lower side	82.85% 57.19%

Notice that flow transitions considerably quicker on the lower side of the capsule, due, perhaps, to the spillover effect. However, this could equally be due

to the poor capturing of the gap BL, making transition results in the air gap unreliable. Although the transportation system presented here is low-drag, silent and emission-free, none of these help when it comes to the large tube/tunneling costs shown in Figure 7. A considerable amount of R&D should go into addressing the challenges encountered in designing a compressor, or a similar system, to help push the allowable blockage ratio up and reduce costs. Another issue which should be investigated is the maximum temperature achieved on the lower side of the capsule, to validate or disprove the results obtained in the present study, since such high temperatures would lead to large thermal stresses, and catastrophic failure if unaccounted for.

Conclusions

To conclude, current ground transportation systems have been reviewed and in order to provide a contender to regional jets, an evacuated tube concept needs to be adopted. The main design drivers have been identified as: Mach number, blockage ratio and tube pressure, and their impact has been assessed. It has been identified that, at the design Mach number ($M = 0.482$), the flow choking of a capsule travelling through an evacuated tube is very similar to regular subsonic channel flow choking, and the acceptable blockage ratio (with viscous corrections) has been identified as $\tau = 0.265$, but to provide very similar performance on cold days and boost system reliability, $\tau = 0.249$ was chosen. The cost reductions associated with having a compressor and increasing the blockage ratio have been presented. Transition has been investigated and the expected mechanism has been identified as a LSB formed under an APG on the tail. The difference in drag values between turbulent and transitional flow regimes is presented in Table 2, while the shapes and fineness ratios for nose and tail were optimised and can be seen in Table 5. Finally a three dimensional study illustrated the main flow characteristics and the relevant forces and moments on the capsule, while a comparison against the 2D predictions is shown in Table 4. Possible future work routes have been identified in: exploring 3D optimisation, investigating the temperatures in the air gap and developing a solution for the complexity issue of a compressor.

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Appendix A: Initial Numerical Setup

The initial axisymmetric study was performed on a computational domain of the type shown in Figure 13.

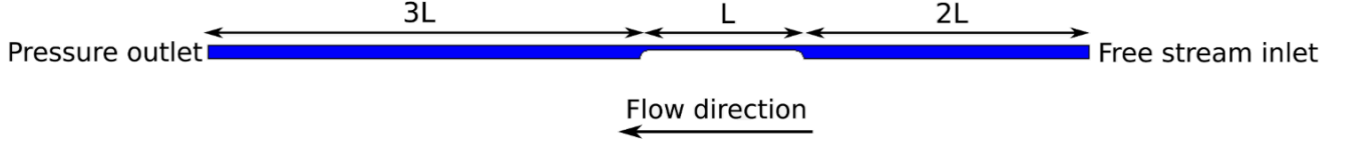


Figure 13: Typical computational domain employed in preliminary study. In this case $L = 22.4\text{m}$.

Two body lengths were kept between the inlet and the body and three body lengths between the body and the outlet. The size of the computational domain had no major impact on the drag estimates as long as the flow was not choked, and this was verified by progressively increasing the size of the computational domain. The choice of boundary conditions is as follows: for lower wall we have axis of revolution to model a 3D body, top wall is a moving viscous wall in accordance with the relative frame of reference, the outlet is simply a pressure outlet, while the inlet is a free stream. One might argue that the flow characteristics should closer resemble those of internal flows, and hence a stagnation inlet B.C. would be more suitable. We reasoned, however, that the setup is very similar to that of a body being tested in a wind tunnel (moving wall or not) and in such situations the common numerical practice is to model the inlet as a free stream B.C. ([24]). The mesh was progressively refined around the body and this is shown in Figure 14. The mesh was carefully designed such as to ensure that the wall y^+ values of both the capsule and the tube were between 0 and 1 and hence that the boundary layers were captured down to (including) the viscous sublayers. The realisable $k - \varepsilon$ turbulence model employed at this stage required two values to be specified at the inlet: turbulent mixing length l_m and turbulence intensity Tu . The mixing length was estimated using ([35]):

$$l_m \approx 0.07D_H \quad (5)$$

where $D_H = D_{tube} - D_{capsule}$. The turbulence intensity was estimated for a fully developed pipe flow ([20]):

$$Tu \approx 0.16Re_H^{-\frac{1}{8}}. \quad (6)$$



Figure 14: Typical mesh employed in preliminary study.

Appendix B: Development of Longitudinal Vortices

In this section we aim to give a clearer image of how the observed longitudinal vortices along the side of the vehicle are formed, by looking at crossflow velocity vectors (in the $\mathbf{j}$ and $\mathbf{k}$ directions). The $\mathbf{j}$ component velocity vectors are shown in Figure 15, and the $\mathbf{k}$ component velocity vectors in Figure 16. The current images were taken for the 10cm gap where the effect of the longitudinal vortices was most pronounced. It can be noticed in both sets of Figures, that starting from $x = 10\text{m}$ air is pushed out of the small gap and that leads to the formation of the longitudinal vortices, which are shown using the λ_2 criterion ([36]) in Figure 10.

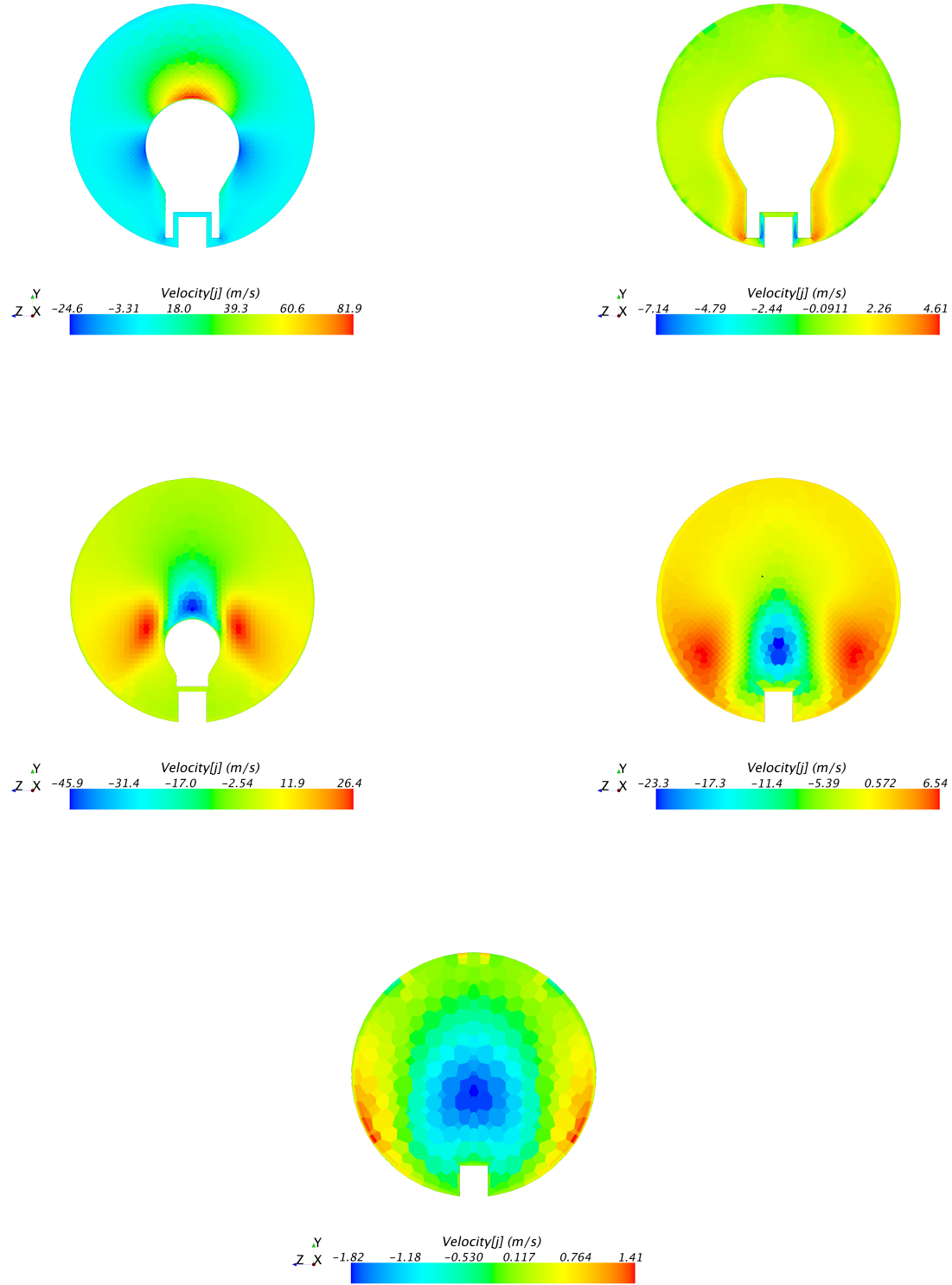


Figure 15: Crossflow velocity vectors (j component) at the following streamwise locations: $x = 2.5\text{m}$, $x = 10\text{m}$, $x = 31\text{m}$, $x = 40\text{m}$, $x = 80\text{m}$, measured from an origin set at the nose tip.

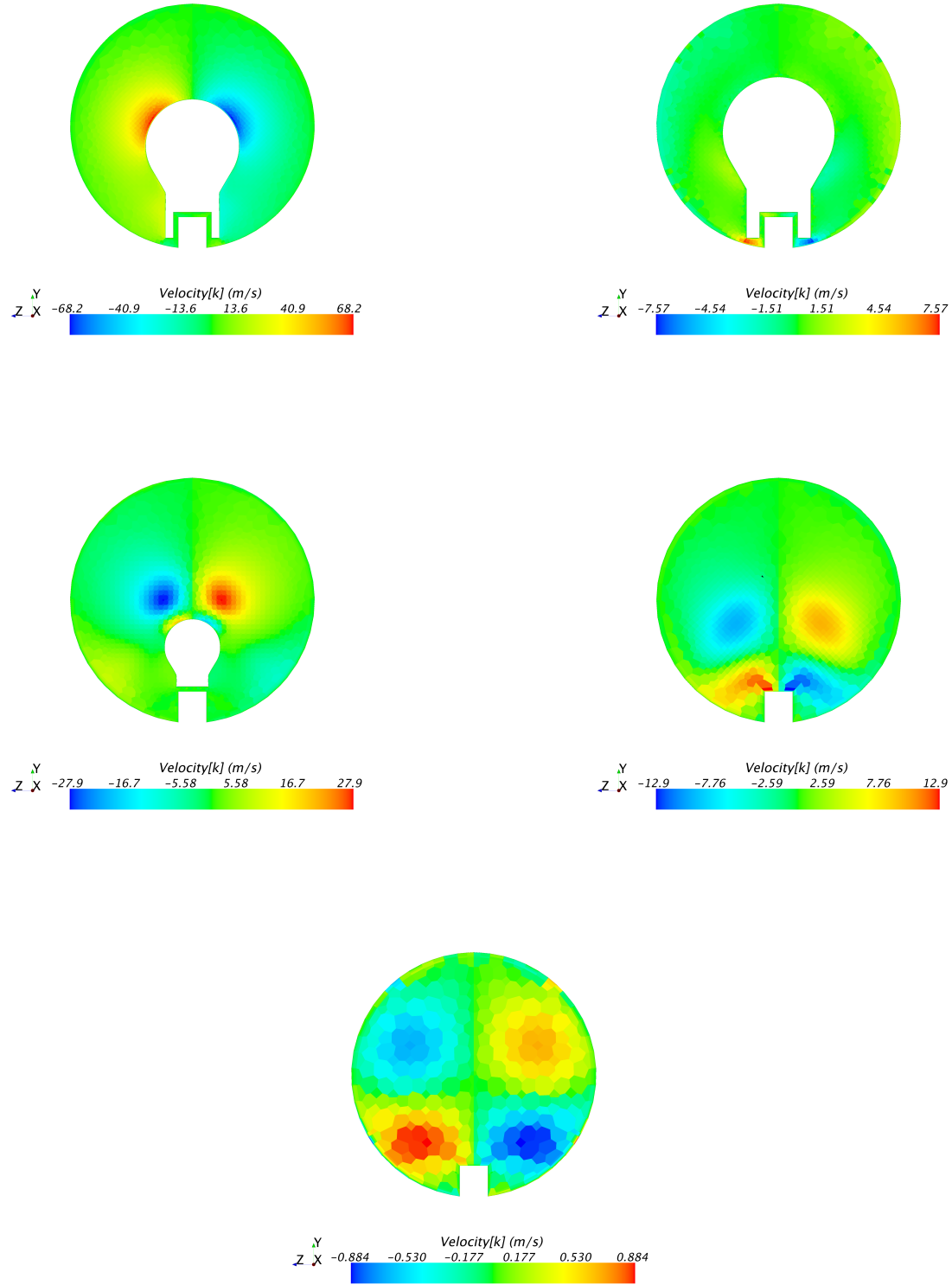
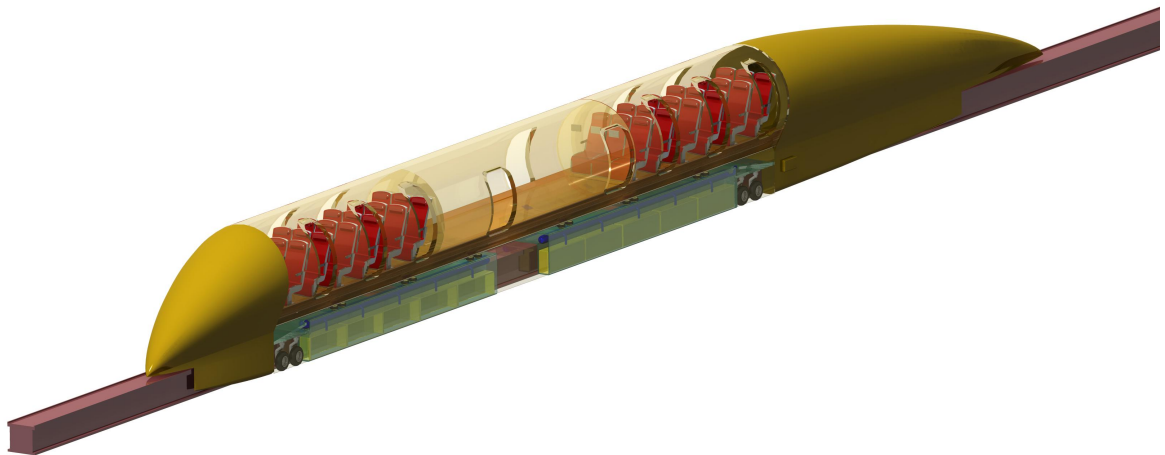


Figure 16: Crossflow velocity vectors ($\mathbf{k}$ component) at the following streamwise locations: $x = 2.5\text{m}$, $x = 10\text{m}$, $x = 31\text{m}$, $x = 40\text{m}$, $x = 80\text{m}$, measured from an origin set at the nose tip.

Appendix C: 3D Aerodynamic Shape

The two different tail positions which are compared against in the 3D aerodynamic optimisation are shown here. The tail down design is highlighted in Figure 17, while the tail up (axisymmetric) design is presented in Figure 18.



Creo Parametric - Advanced Rendering Extension

Figure 17: Tail down 3D aerodynamic design of the capsule. Full assembly rendering.



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Figure 18: Tail up (axisymmetric) 3D aerodynamic design of the capsule. Full assembly rendering.

Appendix D: 3D Meshing

Meshing such complex geometries posed serious difficulties and was much more time consuming than we expected. For most cases the meshes achieved were of good quality with a satisfactory capturing of the boundary layers ($0 < y^+ < 1$), even in the 10cm gap between the track and the underside of the capsule, both BLs were resolved to a satisfactory extent. However, for the 3.5cm gap this task proved difficult. Although the mesh overall was good with a very good resolution of the BLs, in the small gap between the capsule and the track it was impossible to put prism layers (for BL capturing) without achieving very poor cell volume ratios, or a much too fine mesh. This resulted in y^+ values of around 5 or 6 in the gap, which is undesirable and can affect the temperature inside the gap due to a poor capturing of the thermal boundary layer, and also affect the transition point on the lower side of the actual capsule, due to the strict requirements of the $\gamma - Re_\theta$ model regarding BL capturing.

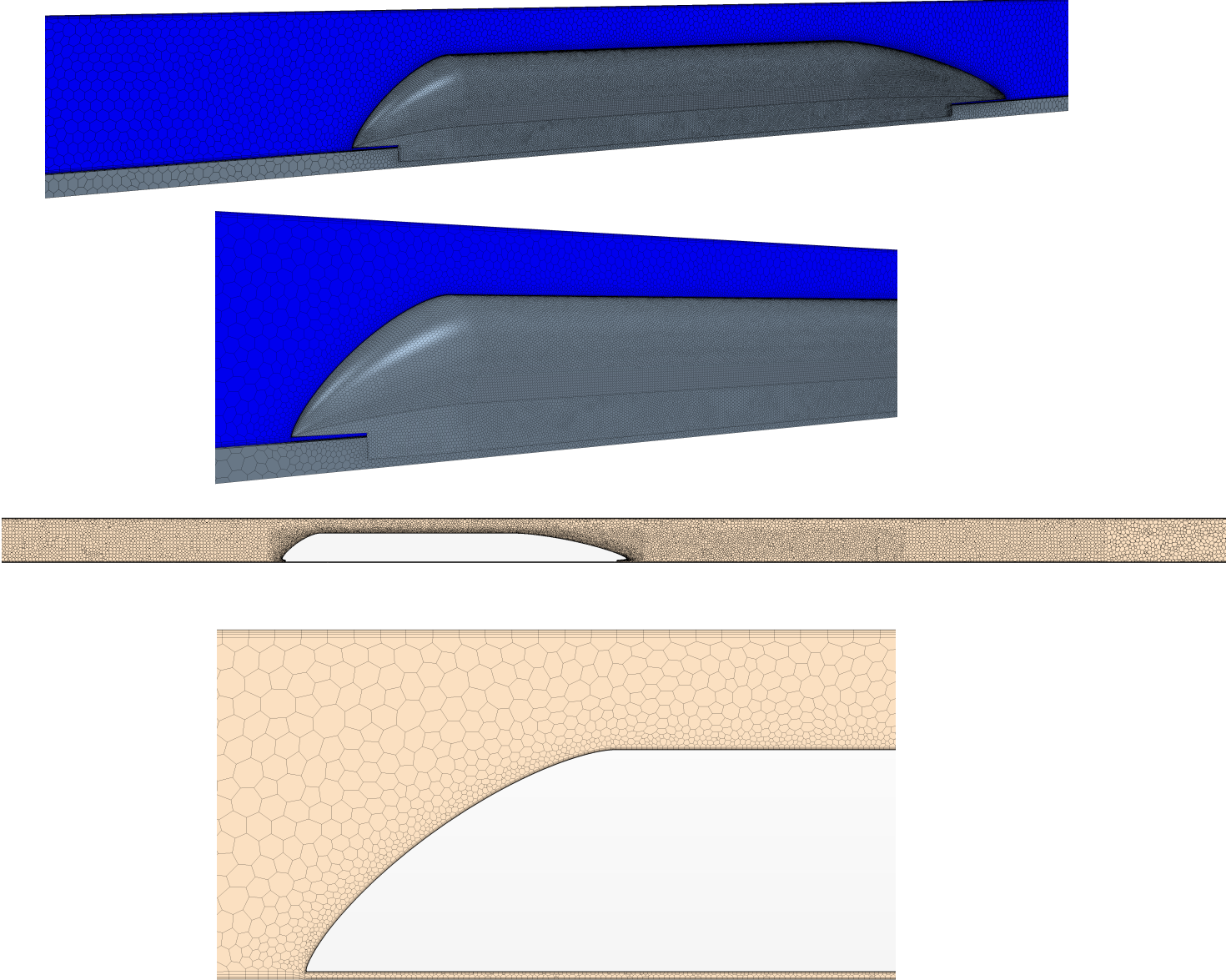


Figure 19: Different views of the mesh for the 10cm gap tail down configuration.